concept that he, Zwicky, had invented to explain the origins of the most energetic phenomena seen by astronomers: supernovae, and cosmic rays. He even went on the air in a nationally broadcast radio show to popularize his neutron stars. But under close scrutiny, his articles and lectures were unconvincing. They contained little substantiation for his ideas.

It was rumored that Robert Millikan (the man who had built Caltech into a powerhouse among science institutions), when asked in the midst of all this hoopla why he kept Zwicky at Caltech, replied that it just might turn out that some of Zwicky's far-out ideas were right. Millikan, unlike some others in the science establishment, must have seen hints of Zwicky's intuitive genius—a genius that became widely recognized only thirty-five years later, when observational astronomers discovered real neutron stars in the sky and verified some of Zwicky's extravagant claims about them.



Fritz Zwicky among a gathering of scientists at Caltech in 1931. Also in the photograph are Richard Tolman (who will be an important figure later in this chapter), Robert Millikan, and Albert Einstein. [Courtesy of the Archives, California Institute of Technology.]

Among Zwicky's claims, the most relevant to this book is the role of neutron stars as stellar corpses. As we shall see, a normal star that is too massive to die a white-dwarf death may die a neutron-star death instead. If *all* massive stars were to die that way, then the Universe would be saved from the most outrageous of hypothesized stellar corpses: black holes. With